

Research on UAV Online Visual Tracking Algorithm based on YOLOv5 and FlowNet2 for Apple Yield Inspection*

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Abstract— Accurate monitoring of fruit quantity in apple orchards will allow growers to manage their orchards more efficiently, leading to higher yields. In addition, the rapid and accurate inspection of fruit in the orchard is also one of the basic technologies needed for smart agriculture. This paper proposes a real-time method for apple tracking and yield estimation. UAV carrying RGB camera is used as an inspection platform, which analyzes the video in real time during the inspection. The algorithm is built according to the Tracking-by-Detecting framework, where YOLOV5 target detection model is used to obtain apples' exact position. Meanwhile, the apples' position in next frame is estimated according to the optical flow calculated from FlowNet2 model. Then, the predicted position and detected position is matched by the Hungarian algorithm. A dataset close to the actual situation in the orchard is constructed to verify the effectiveness of the proposed method. To improve the detection accuracy of apple under actual scene, the data enhancement strategy of random occlusion and mosaic enhancement is used for model training. As a result, the accuracy of apple detection achieved by the algorithm in this paper is 85.5%, which is 11% higher than previous studies. Besides, it can keep well tracking of detected apples even under the influence of complex occlusion or other factors, and achieve an accuracy of 90.39% in apple yield estimation. More importantly, this algorithm can reach a speed of 20FPS on the experimental platform, which meets the real-time requirements of UAV inspection.

I. INTRODUCTION

Nowadays, apple is one of the most widely grown and sold fruits all over the world. China accounts for nearly half of the worldwide apple production according to the data from the Food and Agriculture Organization of the United Nations. However, apple cultivation in China is still dominated by manual labor, with low level of mechanization and automation [1]. With the aging of the population and the migration of labor to cities and towns, China's apple industry urgently needs to be automated and intelligent. Among them, the

regular visual inspection of apple orchard with flying inspection robot and artificial intelligence technology is particularly important, which can provide farmers with the accurate information of individual fruit and yield information of the whole orchard, so that the subsequent orchard intelligent management and apple picking robot precision picking technology can be realized. Among them, apple detection, tracking and counting methods based on machine vision is the key technology. Relevant researches have been carried out and applied in apple pruning [2], yield estimation [3-4], harvesting [5] and other aspects.

In terms of apple's visual detection, early studies mainly used manual extracted image features combined with machine learning methods for fruit detection [6]. However, recent studies have shown that deep learning-based methods can achieve better display results. Inkyu Sa et. al. [7] fused RGB image and NIR image, and then adopted Faster R-CNN model to realize the detection of fruits, which provides a method to merge multispectral data. LJ Biffi et. al. [8] used Adaptive Training Sample Selection (ATSS) deep learning method to reduce the cost of data annotation. X Wang et. al. [9] designed an apple detection network for apple harvesting robot and used the method of instance segmentation to detect apples. K Zhang et. al. [10] also designed the apple instance segmentation network for apple picking robot, and fused the detection result from two cameras. Other recent studies are not too detailed here [11-13]. The above research is aimed at apple detection under the ideal environment of close distance and no serious occlusion, and most algorithms cannot run in real time on the inspection robot. However, what flying inspection robot needs to face is real-time fruit detection for small targets under the condition of serious illumination and occlusion influence. And there are few studies concern on this issue.

Besides, in terms of apple's visual tracking and counting, researchers mainly focus on off-line algorithms. Nicolai Hani et. al. [4] simplified the problem into two parts: detection of apple clusters and classification of apple clusters by number of apples in the cluster, and the algorithm could only calculate the number of apples in a single image. Xu Liu et. al. [14] obtained the apple center by means of semantic segmentation, and realized apple matching in continuous frames through 3d reconstruction of apples, so as to realize apple counting, which is time-consuming. Uddhav Bhattacharya et. al. [15] designed a network based on VGG16 to estimate the number of apples directly from a single image. This method also cannot be applied to apple counts in video sequences. Similar to apple's detection, methods suitable for flying inspection robots are rarely studied.

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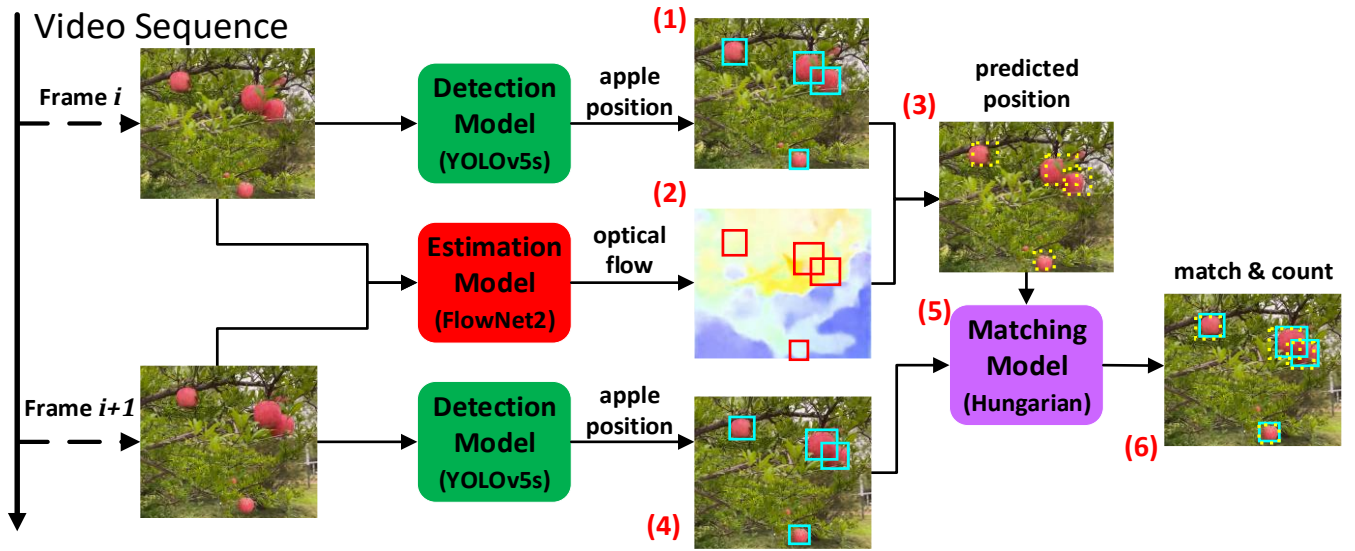


Figure 1. Overall Algorithm Architecture. Continuous frames are taken out as the input of the algorithm from video sequence. (1) The position of apples in frame i is obtained through the YOLOv5s6 detection model; (2) Combining frame i and $i+1$, the information of image motion is obtained by estimation model based on FlowNet2; (3) Apple's position in frame $i+1$ is predicted with the optical flow corresponding to the position of apple; (4) The position of apple in frame $i+1$ is determined by detection model; (5) The predicted position of apple in frame $i+1$ is matched with apple's actual position; (6) Number of apple is updated by counting strategy.

This paper aims to provide a real-time visual tracking algorithm used for apple yield inspection with UAV. As shown in Fig.1, the algorithm includes YOLOv5 apple detection module, position estimation module based on FlowNet2 and matching counting module based on Hungarian algorithm. Combined with the UAV inspection method proposed in this paper, the apple yield in the orchard can be obtained in real time during the inspection. The main contributions of our paper are: (1) a flying inspection method for apple orchard; (2) a real-time apple visual tracking algorithm used for apple quality inspection; and (3) a strategy for counting both healthy and diseased apples based on Hungarian matching algorithm.

II. PROPOSED METHOD

A. UAV Inspection method

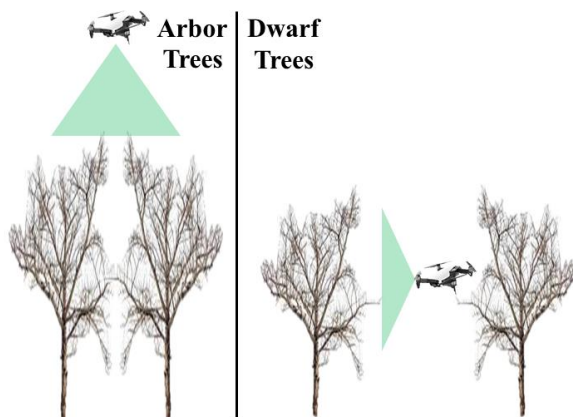


Figure 2. UAV inspection method in apple orchard

In order to improve the efficiency of inspection, UAVs carrying RGB cameras are used as inspection platforms. The inspection method of UAV determines the characteristics of the images obtained during inspection. Apple trees in orchards are usually planted in rows. Therefore, the UAV inspection method designed in this paper is shown in Fig. 2. For tall and densely planted arbor trees, UAV will take images from the top down for inspection; for dwarf trees with large planting width, the UAV will fly over the middle of the tree path and take images from the horizontal direction for inspection.

Therefore, in the images obtained by inspection, fruits are usually small and seriously blocked by leaves and other objects, accompanied by strong light changes. In addition, due to the high flight speed of UAV, the target movement distance between frames in the inspection video sequence is large. Therefore, the apple tracking algorithm designed in this paper based on the framework of tracking-by-Detecting. And the problem about apple's yield inspection is divided into three sub-problems: (1) Visual detection of small sized apples that are shaded and affected by light and shadow (2) Position estimation of detected apples in consecutive video frames (3) Matching between estimated position and detected position of apples followed by the counting of both kind of apples. The detailed algorithm architecture is shown in Fig. 1. The input of the algorithm is real-time images taken by the UAV.

B. Detection with YOLOv5s

Compared with traditional machine vision methods, the object detection method based on deep learning has more advantages in accuracy and generalization ability. They are further divided into one-stage and two-stage, among which the method detection speed of one-stage is faster. Therefore, this method is more suitable for real-time detection application

scenarios. YOLO series of algorithms are representative of one-stage class detection methods, which are widely used in various target detection tasks. Fan et. al. [16] applied the pruned YOLOv4 model to the apple detection task on the apple sorting line, and obtained an average accuracy of 93.9%.

As an upgraded version of YOLOv4, YOLOv5 is widely used by researchers for its advanced detection speed and accuracy. Its improvement was reflected in three aspects: (1) Mosaic data augmentation, adaptive image scaling and other image preprocessing methods were adopted, which greatly optimized the performance of the model; (2) CSPNet is used in backbone and head to strengthen the ability of network feature fusion; (3) The structure of FPN and PANet in head accelerates the information flow between various levels. According to the results presented by the authors, YOLOv5s can run at a speed of 6.4ms per frame on COCO dataset, which shows the great real-time performance.

In order to complete the accurate detection of apple in the complex environment and ensure the real-time direction of the algorithm, the lightweight YOLOv5s model is used as the core module of the detection. The S model of YOLOv5 also has several versions, the most commonly used ones are YOLOv5S and YOLOv5S6. On the basis of the former, the latter uses convolution to replace the Focus structure, and uses CSPNet containing three layers of convolution to replace various types of CSPNet in original model, which increases the depth of the network and the number of parameters. The detection accuracy is improved at the expense of running speed. In the training stage of the model, the transfer learning strategy is applied to accelerate the model training speed. The parameters of the YOLOv5s model are initialized by its pre-trained model on the COCO dataset.

C. Position Estimation Based on Optical Flow

Under the framework of tracking-by-detecting, the core of target tracking lies in the matching of corresponding objects between successive frames. Predicting the target's possible position in the next frame can greatly improve the accuracy of matching. In multi-target visual tracking, Kalman filter, particle filter, optical flow and other methods are often used to estimate targets' position [17]. Since video images may contain many apples, it is not practical to build tracking models for each individual. Therefore, it is a more feasible scheme to directly calculate pixel-level image motion information by optical flow method and then obtain the estimation of apples' position.

Studies in recent years show that optical flow diagrams obtained by deep learning-based methods have higher quality and faster inference speed compared with traditional methods such as Farneback [18]. In the experiment of this paper, the traditional method takes about 100ms for each frame of optical flow, which cannot meet the real-time requirements of the algorithm. FlowNet2 is a deep learning optical flow method with good performance at present [19]. It continuously optimizes the optical flow diagram by stacking FlowNetC and FlowNetS on the basis of FlowNet. In order to

ensure the real-time performance of the tracking algorithm, the FlowNet2CS module, part of FlowNet2, is selected as the reasoning model of optical flow.

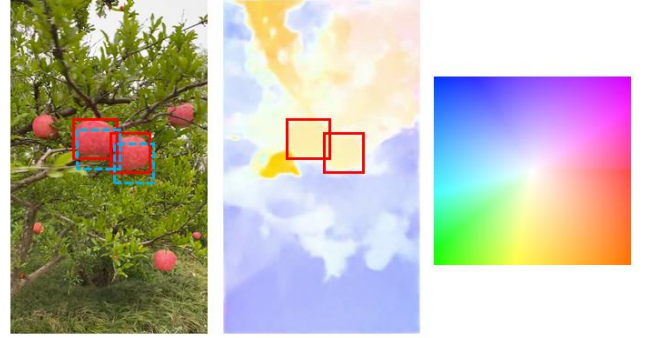


Figure 3. Apples' position estimation based on optical flow

The process of apple position estimation based on this algorithm is shown in Fig. 3. In the figure, the detected bounding box is in red, and apple's bounding box in blue is estimated in the next frame. The left picture is the RGB image taken in inspection, and the middle one is the visual optical flow diagram, where the color represents the direction of pixel movement. The right picture is the legend of the visual optical flow diagram. Since the optical flow of a single pixel has certain noise, the average value of all optical flows in apple's bounding box (red box) is regarded as the movement information of the center points of the bounding box. The final estimated position of the apple is determined by the following formula.

$$F_{ave}^i = \frac{1}{h^i \times w^i} \sum F(x^i, y^i) \quad (1)$$

$$(x^i, y^i) \in bbox^i \quad (2)$$

$$\hat{P}_{t+1}^i = P_t^i + F_{ave}^i \quad (3)$$

where: $bbox^i$ is the i th apple's bounding box in the image. $F(x, y)$ is the optical flow at (x, y) . h^i and w^i are the height and width pixel values of the $bbox^i$ respectively. F_{ave}^i is the average optical flow of pixels in $bbox^i$. P_t^i and $\hat{P}_{t+1}^i$ are respectively the position of $bbox^i$ at the current moment and its predicted position at the next moment.

D. Object Matching and Counting

As the final step in tracking, the predicted apple locations need to be matched with the actual detected apple locations to track the instance. The core of matching is to find the most matching pairs from all possible matching relationships.

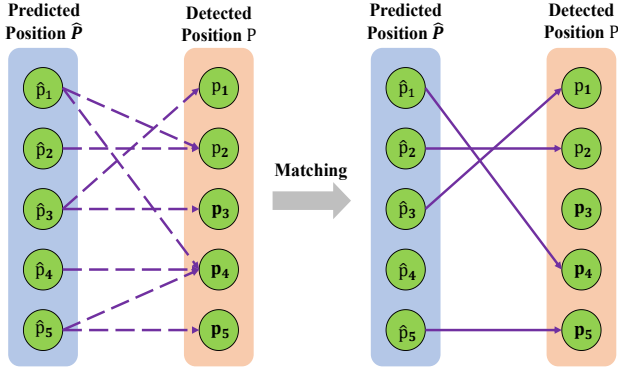


Figure 4. Bipartite graph model of apple matching. Dotted lines represent possible matching relationships, and solid lines represent the final matching result.

As shown in Fig. 4, the problem can be described by the bipartite graph model, described as (4) and (5). In the process of tracking, an apple may be blocked for a certain period of time, and its position and pose can only be estimated by optical flow method. When the apple is detected again, its estimated position has a large deviation from the actual position. Therefore, as shown in (6)(7)(8), the judgment conditions of potential matching pairs consist of two points : (a) the IOU of predicted bounding box and detected bounding box is greater than the given threshold; (b) The prediction bounding box and the detection bounding box do not intersect but the distance is less than the given threshold. The mathematical description of this problem is shown in the following formula.

$$G = (V, E) \quad (4)$$

$$V = \hat{P} \cup P, \quad \hat{P} \cap P = \emptyset \quad (5)$$

$$\forall (\hat{p}, p) \quad \hat{p} \in \hat{P}, p \in P \in E, \begin{cases} IOU_{\hat{p}p} \geq 80\% , IOU_{\hat{p}p} > 0 \\ Dis_{\hat{p}p} \leq 30 , IOU_{\hat{p}p} = 0 \end{cases} \quad (6)$$

$$IOU_{\hat{p}p} = \frac{bbox_{\hat{p}} \cap bbox_p}{bbox_{\hat{p}} \cup bbox_p} \quad (7)$$

$$Dis_{\hat{p}p} = \|center_{\hat{p}} - center_p\|_2 \quad (8)$$

where: G is a binary graph composed of node V and edge E . Node V is divided into two disjoint subsets, namely, the predicted position $\hat{P}$ and the actual detected position P . $center_p$ is the center of bounding box of position p .

In the binary graph model described above, our goal is to find as many matches as possible. The Hungarian algorithm is usually used to solve the maximum matching problem of binary graphs. Therefore, the Hungarian algorithm is used to obtain the matching relationship between apple's estimated position and the new detected position. In the matching results, it is possible that some apples are not matched, as shown in

Fig. 6. The apple that is not matched in the predicted position means that the apple is not detected in the new frame, and the apple that is not matched in the detected position is the new apple in the image. To avoid double counting, the counting strategy is designed as follows:

1. Open a video sequence and detect the apples in first frame to initialize the count.
2. Read a new frame from the video sequence.
3. Calculate the optical flow and estimate the position of detected apples.
4. Detect the apples in new frame.
5. Matches detected apples with newly detected apples
6. For the matched individual in detected apple, the location is updated based on matching results.
7. For the unmatched individual in the detected apple, the actual position is updated with the estimated position. If the unmatched individual does not appear in 10 consecutive frames or the detection frame is beyond the range of the image, the individual is deleted.
8. For unmatched individuals in newly detected apples, add them to the current apple queue and update the count.
9. Go to 2 if the video does not end, otherwise save the video processing result and end the program.

III. EXPERIMENT AND RESULT

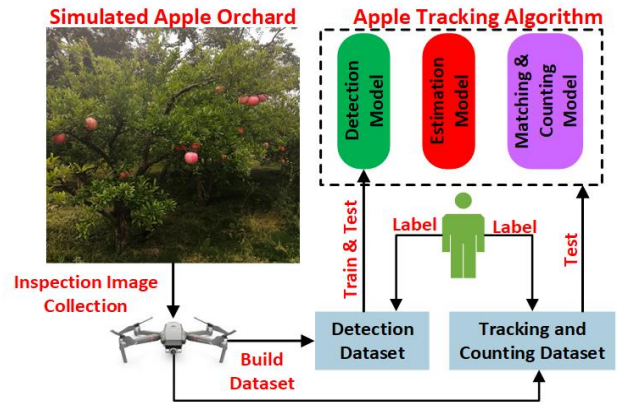


Figure 5. Experimental principle

The experimental principle is shown in Fig. 5. Model apples were hung on trees to simulate real orchards. Then, the orchard inspection images and videos obtained by UAV are used to construct apple detection dataset and apple tracking and counting data set respectively. Manually marked apple detection dataset is used for training and testing the apple detection module. The apple tracking and counting dataset is used to verify the performance of the whole algorithm. The experiment is carried out on NVIDIA GeForce 940MX platform with only 2GB video memory, which can well simulate the computing performance of embedded platform.

A. Data Acquisition

The data set constructed in this article consists of partial images in the Minneapple dataset [20] as well as images captured in the simulated experiment, of which 62.2% were captured experimentally. The dataset contains 1790 simulated real inspection images for Apple inspection and 9 videos for verification of the Apple tracking and production estimates. In the image, 28084 apples were manually labeled for training and testing the network. The dataset used to detect Apple was split into separate training and testing set, with 1,211 and 579 images, respectively.



Figure 6. Images in dataset

The images in the dataset are shown in the Fig. 6. The left image is the selected Minneapple dataset and the right image is the inspection image collected from the experiment. In the experiment, the apple model was hung on the outdoor fruit tree according to the growth status of apple on the actual fruit tree, and DJI Mavic 3 was used as the inspection platform to shoot inspection images and videos. The data set fully simulates the characteristics of small apple targets, which are seriously affected by occlusion and illumination.

B. Apple Detection

TABLE I. TRAINING RESULTS OF APPLE DETECTION MODEL

Model	Parameters and Performance Results					
	Image Size	Precision	Recall	mAP.5	mAP0.5:0.95	Speed
YOLOv5s	640	0.842	0.783	0.867	0.491	32ms
YOLOv5s	1280	0.853	0.827	0.897	0.531	107ms
YOLOv5s6	640	0.853	0.783	0.864	0.469	35ms
YOLOv5s6	1280	0.855	0.826	0.902	0.534	132ms

The accuracy of apple detection obtained in previous studies on the Minneapple dataset was only 77% [20]. The model adopted in this paper has a significantly improved accuracy of 85%. In order to conduct comparative verification and select the optimal model, this paper selected multiple versions of YOLOv5s and set image input of different sizes for training. The training results of each model are shown in Tab. I. Among them, the detection accuracy of YOLOv5s6 model with an image of 640 size as input is 85.8%, which takes only 32ms, achieving a balance in accuracy and

real-time performance. Therefore, this model is used as the detection module of tracking algorithm.



Figure 7. Results of apple detection

The actual detection results are shown in Fig. 7. It can be seen that the model can achieve more accurate detection for small apples and apples in shadow or shade. The model even detected apples that had not been manually labeled. Therefore, the data in Tab. 1. may underestimate the performance of the model.

C. Apple Tracking and Counting

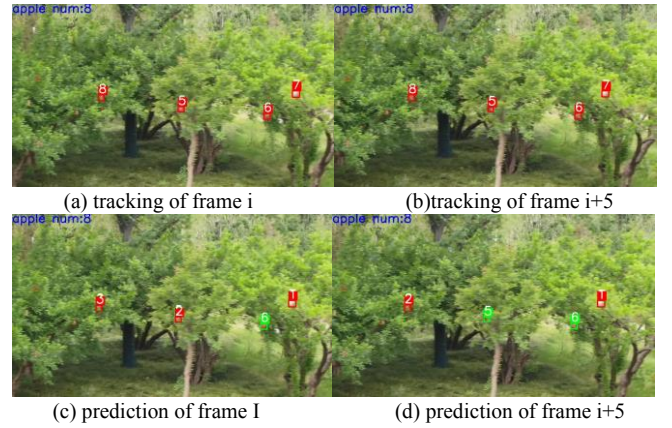


Figure 8. process of the Apple tracking and counting

Tracking and counting of Apple are shown in Fig. 8. (a) and (b) are the tracking and counting results of two frames separated by a period of time in the video sequence respectively, where the red box indicates the tracking target and the number represents the ID of the current individual. (c) and (d) are the corresponding position prediction and matching schematic diagram. In the figure, the red box represents the newly detected target of the detection model, and the green box represents the estimated position of apple. As can be seen from the figure, there is a high degree of coincidence between the estimated location and the actual detected location. Even if the current location of the individual is not updated continuously according to the newly detected location, the prediction model can still track the corresponding individual well, as shown in individual 6.

TABLE II. RESULTS OF APPLE COUNTING

Video number	Result of apple counting		
	Ground truth	Count	Precision
1	29	27	93.10%
2	30	27	90.00%
3	31	29	93.55%
4	20	17	85.00%
5	15	16	93.33%
6	40	45	87.50%
7	33	33	100.00%
8	35	44	74.29%
9	31	32	96.77%
Standard Deviation			7.12%
Average precision			90.39%

Finally, the algorithm is run on 9 videos in the dataset, and the counting results are shown in Tab. II. It can be seen that the algorithm can accurately count the number of apples in the inspection video with an average precision of 90.39%. And the algorithm only takes about 50ms to analyze each frame, which means the algorithm can run at 20FPS on the embedded platform on the UAV to estimate orchard yield in real time during training.

IV. CONCLUSION

A real-time UAV visual tracking algorithm for apple yield inspection is proposed. A dataset suitable for apple orchard flying inspection was constructed to train and verify the algorithm. The YOLOv5s6 model is used as the apple detection model, and the detection accuracy reached 85%. FlowNet2CS model is used to get optical flow for apple location prediction. By building the bipartite graph model, applying Hungarian algorithm and combining with the counting strategy, tracking and counting of apple in the orchard was realized. And the average counting accuracy reached 90.39%. Besides, the whole algorithm can run at the speed of 20FPS, which means this algorithm can work on the UAV in real time.

Future work includes adding apples with diseases and insect pests to the data set, train the algorithm to realize the detection of diseases and insect pests, and finally realize the rapid inspection of apples in the orchard. In addition, this method will be transferred to apple picking robot to achieve precise apple picking.

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